

Discovering Hidden Sector Relationships Among S&P 500 Companies Using Text Analytics and Unsupervised Learning

Tak Hei Wong
ISYE 6740 - Fall 2025

December 1, 2025

Abstract

Market Participants have relied on the Global Industry Classification Standard (GICS) since 1999. However, conglomerates nowadays often operate across several domains, challenging static, and one-dimensional classification systems.

This project constructs text-based representations of S&P 500 companies using Wikipedia articles and evaluates how well interpretable machine learning models can reproduce and challenge existing sector definitions. We start by applying TF-IDF vectorization with sublinear term frequency scaling, dimensionality reduction via Truncated SVD, K-Means and Spectral clustering, and followed by supervised classification (Logistic Regression and Linear SVM). In our experiments, Linear SVM achieves 85.9% cross-validated accuracy predicting GICS sectors. Spectral clustering on the other hand, achieves the best unsupervised performance (ARI: 0.439, NMI: 0.626), capturing 63% of GICS structure while revealing cross-sector patterns. The uniformly low silhouette scores across all cluster numbers also suggest that companies exist in continuous operational space rather than a discrete one, motivating potential applications in alternative factor-based portfolio construction.

1 Introduction

1.1 Motivation

The Global Industry Classification Standard (GICS) [S&P Dow Jones Indices, 2025, MSCI, 2025], established in 1999, classifies companies into 11 major sectors. However, modern corporations such as Amazon, Microsoft, and Google increasingly operate across multiple domains, challenging these static boundaries.

1.2 Research Questions

This project addresses the following topics:

1. **Reproduction of GICS sector from Wikipedia articles features:** We evaluate whether publicly available textual descriptions contain sufficient information to predict official sector labels.
2. **Discovery of hidden structures in company descriptions beyond GICS:** Using unsupervised clustering, we identify groupings that might uncover themes or emerging industry categories across the different sectors.
3. **Companies outlier:** By analyzing misclassifications and mixed clusters, we identify firms that exhibit characteristics of multiple sectors.

1.3 Related Work

Text-based company classification has been explored using various approaches. Classical methods apply TF-IDF with SVM and Naive Bayes for document classification [Allahyari et al., 2017, Lamby and Iseman, 2018], while recent work leverages transformer models like FinBERT [Huang et al., 2023] and achieves higher accuracy. Alternative approaches use SEC 10-K filings for dynamic industry classification [Hoberg and Phillips, 2016, Winton Research, 2018]. Text-based factors have also been explored for portfolio construction [Bybee et al., 2023].

Our work differs from the abovementioned studies in several ways. First, we utilized Wikipedia articles, making the approach applicable to public companies and private companies alike. While Marti [2023] used deep sentence embeddings, we prioritize interpretability through the use of TF-IDF, achieving competitive accuracy (85.9% cross-validated) while providing explicit keyword-based explanations.

Our methodology combines TF-IDF vectorization, Truncated SVD for dimensionality reduction, K-Means and Spectral clustering for unsupervised analysis, and Logistic Regression and Linear SVM for supervised classification with 5-fold cross-validation.

2 Data Collection and Preprocessing

2.1 Data Sources

We collected data from publicly available sources:

1. **S&P 500 Constituent List:** The Wikipedia page “List of S&P 500 Companies” provides company names, stock tickers, and GICS sector classifications. We used Python’s Pandas library to read and parse the constituents table and extract Wikipedia URLs directly from hyperlinks, yielding 502 stock entries as of November 2025. After filtering out 2 companies without Wikipedia articles (Insulet Corporation and Qnity Electronics, a newly added company as of November 2025) and removing 3 duplicate share classes (Class A and B shares for Alphabet Inc., Fox Corporation, and News Corp), we analyzed 497 unique companies.
2. **Wikipedia Articles:** For each company, we retrieved its full Wikipedia article using the `wikipediaapi` Python library. The article act as our source of comprehensive, publicly-edited descriptions of company operations, products, and services.

2.2 Text Filtering

To focus on current business operations rather than historical context, or backstory about the companies, we implemented some section filtering criteria:

- **Sections excluded:** History, Historical Background, Founding, Timeline, Legacy, Controversies, References, External Links
- **Sections retained:** Summary, Products and Services, Business Model, Operations, Technology, Revenue

2.3 Text Preprocessing

We applied standard NLP data preprocessing steps to enhance our model performance:

1. **Lowercasing:** Normalize all text to lowercase
2. **Punctuation Removal:** Strip special characters using regex
3. **Stop Word Removal:** Remove common English words with scikit-learn’s built-in list
4. **Whitespace Normalization:** Collapse multiple spaces into one

Figure 1 illustrates these transformations using Apple Inc. as an example.

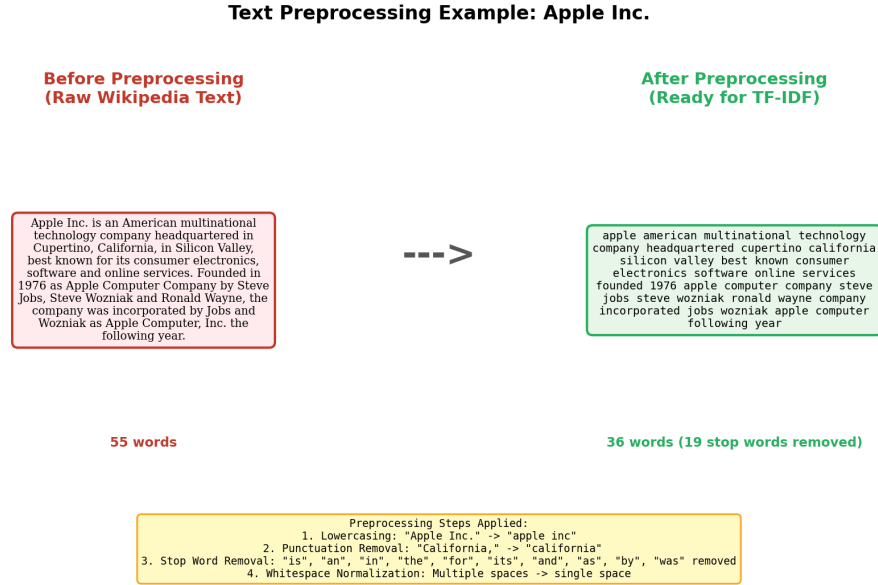


Figure 1: Text preprocessing example for Apple Inc. Left: Raw Wikipedia text (55 words). Right: Fully preprocessed text (36 words) ready for TF-IDF vectorization. The pipeline reduces word count by 35% through: (1) lowercasing (“Apple Inc.” → “apple inc”), (2) punctuation removal (“California,” → “california”), (3) stop word removal (19 common words like “is”, “an”, “the”, “for”, “and” removed), and finally (4) whitespace normalization.

2.4 Data Statistics

Table 1 presents summary statistics of the collected dataset.

Table 1: Dataset Statistics

Metric	Value
S&P 500 constituent table entries	502
Entries without Wikipedia articles	2 (PODD, Q)
Entries with Wikipedia articles	500
Duplicate share classes removed	3 (GOOG, FOX, NWS)
Unique companies analyzed	497
Number of GICS sectors	11
Average article length (chars)	14,903
Median article length (chars)	9,780
TF-IDF vocabulary size	5,000

We successfully retrieved all 497 articles, with an average article length of 14,903 characters (approximately 2,448 words), providing substantial textual content for feature extraction.

3 Methodology

3.1 Feature Extraction: TF-IDF

We convert text documents into numerical feature vectors using Term Frequency-Inverse Document Frequency (TF-IDF) [Salton and Buckley, 1988, Manning et al., 2008]:

$$\text{TF-IDF}(t, d) = \text{TF}(t, d) \times \text{IDF}(t) \quad (1)$$

where:

$$\text{IDF}(t) = \log \frac{N}{|\{d : t \in d\}|} \quad (2)$$

Parameters:

- `ngram_range=(1, 2)`: Capture unigrams and bigrams
- `max_features=5000`: Limit vocabulary size (captures 89% of word occurrences while reducing memory by 48%)
- `min_df=2`: Term must appear in at least 2 documents
- `max_df=0.8`: Term cannot appear in more than 80% of documents
- `sublinear_tf=True`: Apply sublinear term frequency scaling ($\log(1 + \text{tf})$), which improved cross-validated accuracy by +6.0% compared to baseline

This produces a sparse matrix $X \in \mathbb{R}^{497 \times 5000}$.

3.2 Dimensionality Reduction: Truncated SVD

In order to address the curse of dimensionality and reduce computational complexity, we apply Truncated Singular Value Decomposition (SVD). We opted for Truncated SVD rather than PCA because it works directly with sparse matrices, avoiding the need to center the data—a critical advantage for high-dimensional TF-IDF features.

$$X \approx U_k \Sigma_k V_k^T \quad (3)$$

We set an initial target of retaining 95% of explained variance, dynamically selecting the number of components k to achieve this threshold. However, due to the extreme sparsity of text data, our algorithm selected $k = 300$ components, retaining 78.0% of cumulative variance—the maximum achievable within computational constraints.

Figure 2 illustrates the cumulative variance explained as a function of the number of components.

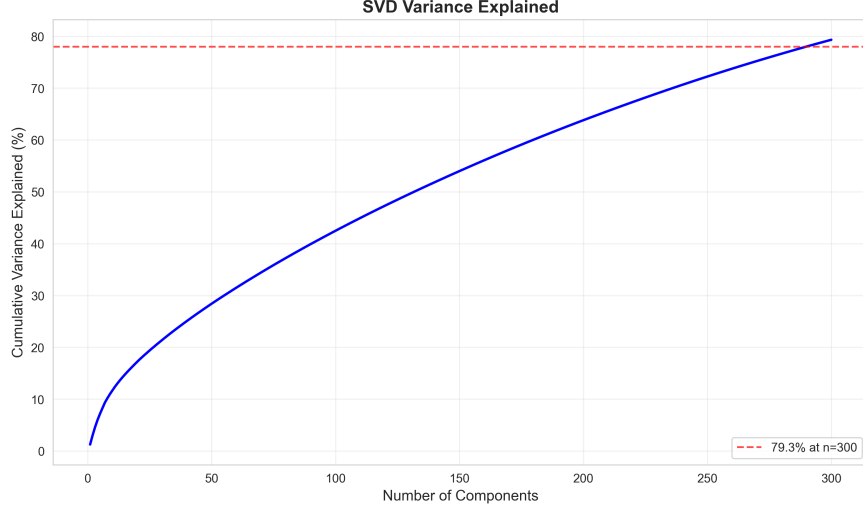


Figure 2: Cumulative variance explained by SVD components. The algorithm selected 300 components, retaining 78.0% of the variance.

3.3 Unsupervised Clustering

3.3.1 Selection of k : Elbow Method and Silhouette Analysis

We set $k = 11$ clusters to match the number of GICS sectors, enabling direct comparison between unsupervised discoveries and established taxonomy. To validate this choice, we performed elbow and silhouette analysis across $k \in \{2, 3, \dots, 20\}$ (Figure 3).

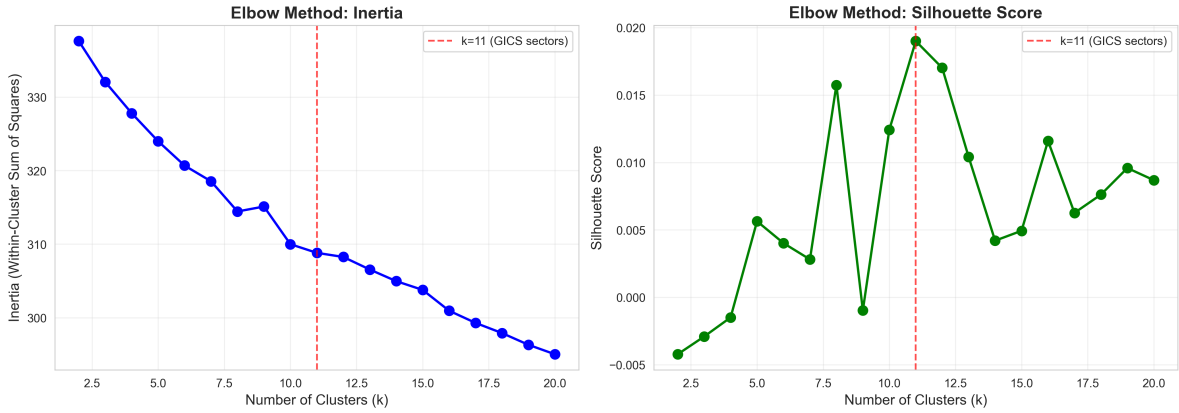


Figure 3: Elbow method and silhouette analysis for selecting optimal k . Left: Inertia decreases continuously with no clear elbow. Right: Silhouette scores remain close to zero across all values of k , indicating overlapping clusters in high-dimensional space. These patterns confirm no natural cluster structure exists, supporting our choice of $k = 11$ for interpretability.

The inertia plot shows continuous decrease without a pronounced elbow, suggesting the data lacks naturally well-separated clusters. The silhouette scores remain uniformly near zero across all values of k , suggesting that companies might not be closer to their assigned cluster centroid than to neighboring clusters. This reveals companies exist along continuous gradients rather than discrete groups—a finding with implications for our further study of factor-based analysis in Section 5.

3.3.2 K-Means Clustering

K-Means [Pedregosa et al., 2011] partitions companies into $k = 11$ clusters by minimizing within-cluster sum of squares:

$$\arg \min_C \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2 \quad (4)$$

where $\mu_i = \frac{1}{|C_i|} \sum_{x \in C_i} x$ is the centroid (mean) of cluster C_i . The algorithm alternates between two steps, (1) assigning data points to nearest centroid, and (2) recomputing centroids, until convergence.

3.3.3 Spectral Clustering

Spectral clustering [Ng et al., 2001] constructs a network of similar companies based on pairwise affinity (nearest-neighbors similarity), then identifies natural groupings by analyzing the connectivity structure of this network through eigenvalue decomposition of the normalized graph Laplacian:

$$L = I - D^{-1/2} A D^{-1/2} \quad (5)$$

where A is the affinity matrix and D is the degree matrix. The first k eigenvectors of L provide a low-dimensional embedding that is then clustered using K-Means. Unlike K-Means which assumes spherical clusters, spectral clustering can capture non-convex cluster shapes and is particularly effective for identifying companies with nuanced, multi-faceted relationships.

3.4 Evaluation Metrics for Clustering

We evaluate clustering quality using both internal and external metrics:

Internal Validation:

- **Silhouette Score:** Measures cohesion and separation, ranging from -1 (poor clustering) to $+1$ (well-separated clusters)

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \quad (6)$$

where $a(i) = \frac{1}{|C_i|-1} \sum_{j \in C_i, j \neq i} \|x^i - x^j\|$ is the mean distance to other points in the same cluster, and $b(i) = \min_{j \neq i} \frac{1}{|C_j|} \sum_{k \in C_j} \|x^i - x^k\|$ is the mean distance to the nearest neighboring cluster.

External Validation (vs. GICS labels):

- **Adjusted Rand Index (ARI)** [Hubert and Arabie, 1985]: Measures agreement between predicted clusters and true labels, adjusted for chance (0 = random agreement, 1 = perfect match). Higher values indicate discovered clusters align well with GICS sectors.
- **Normalized Mutual Information (NMI)** [Strehl and Ghosh, 2002]: Quantifies information shared between cluster assignments and GICS sectors (0 = independent, 1 = identical). Measures how much knowing the cluster assignment tells us about the true GICS sector.

3.5 Supervised Classification

As a benchmark for the informativeness of textual features, we train two classifiers to predict GICS sectors: Logistic Regression, a linear model with interpretable coefficients, and Linear SVM, where its robust to high dimensions).

We employ 5-fold stratified cross-validation for robust performance estimation. Evaluation metrics include:

- **Accuracy:** Measuring overall correct classification rate
- **F1-Score:** A metric incorporating both precision and recall

4 Results

4.1 Dimensionality Reduction

The TF-IDF matrix with 497 companies and 5,000 features was reduced to 300 components, retaining 78.0% of the variance. This dimensionality reduction enables efficient clustering and classification while preserving the majority of information content.

4.2 Clustering Results

4.2.1 Quantitative Evaluation

Table 2 summarizes clustering performance.

Table 2: Clustering Evaluation Metrics

Method	Silhouette	ARI	NMI
K-Means	0.019	0.336	0.586
Spectral	0.026	0.439	0.626

The low Silhouette Scores confirm that companies don’t form distinct, well-separated groups. Instead, they blend together along continuous gradients. Spectral clustering performs better at finding meaningful patterns (ARI: 0.439, NMI: 0.626) than K-Means (ARI: 0.336, NMI: 0.586). This suggests that spectral clustering’s approach of identifying companies with similar neighbors works better for text data than K-Means’ approach of grouping by geometric distance. Both methods capture about 60

4.2.2 Cluster Visualization

Figures 4, 5, and 6 show 2D PCA projections of the clustering results.

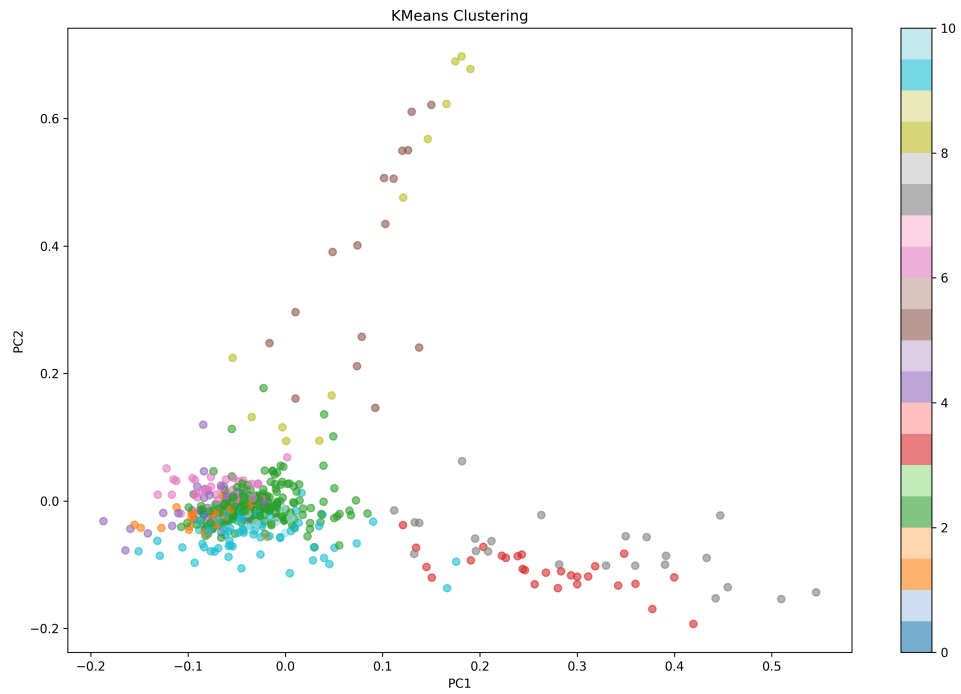


Figure 4: K-Means Clustering visualization (2D PCA projection)

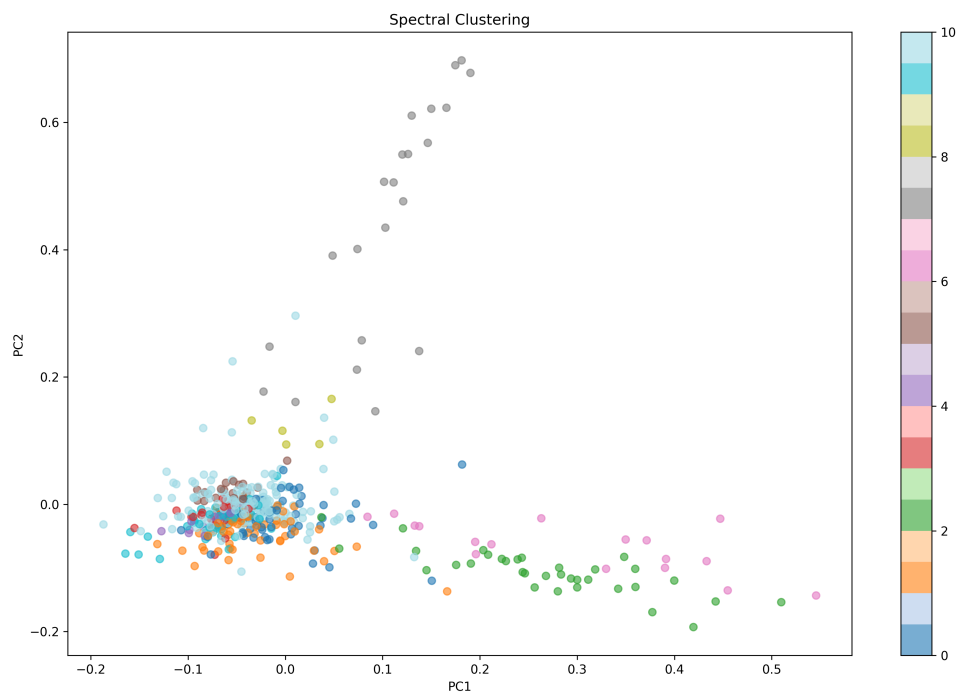


Figure 5: Spectral Clustering visualization (2D PCA projection)

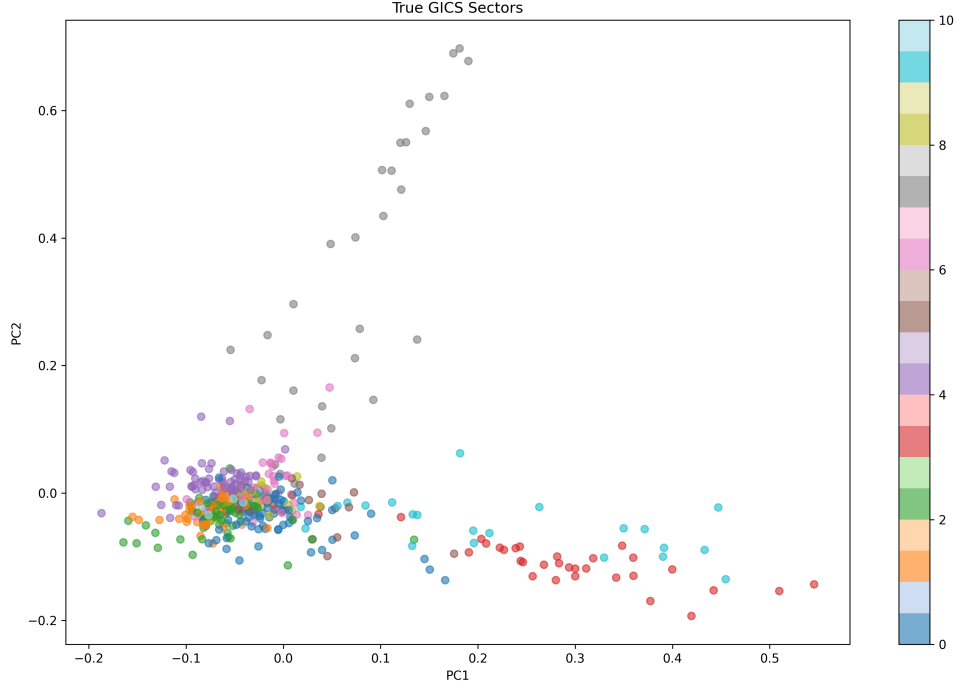


Figure 6: True GICS sector labels (2D PCA projection)

These visualizations reveal fluid boundaries between company groupings rather than discrete clusters. Most companies ($\sim 80\%$) occupy a central region where cluster assignments intermix, with distinct peripheral groups emerging only at the extremes.

This pattern suggests that sector classification exists along a continuous spectrum of business characteristics. Even the true GICS sector plot (Figure 6) shows substantial boundary overlap, indicating that text-based operational descriptions naturally resist categorical separation.

4.2.3 Cluster Composition

Figure 7 presents a heatmap showing the overlap between K-Means clusters and GICS sectors.

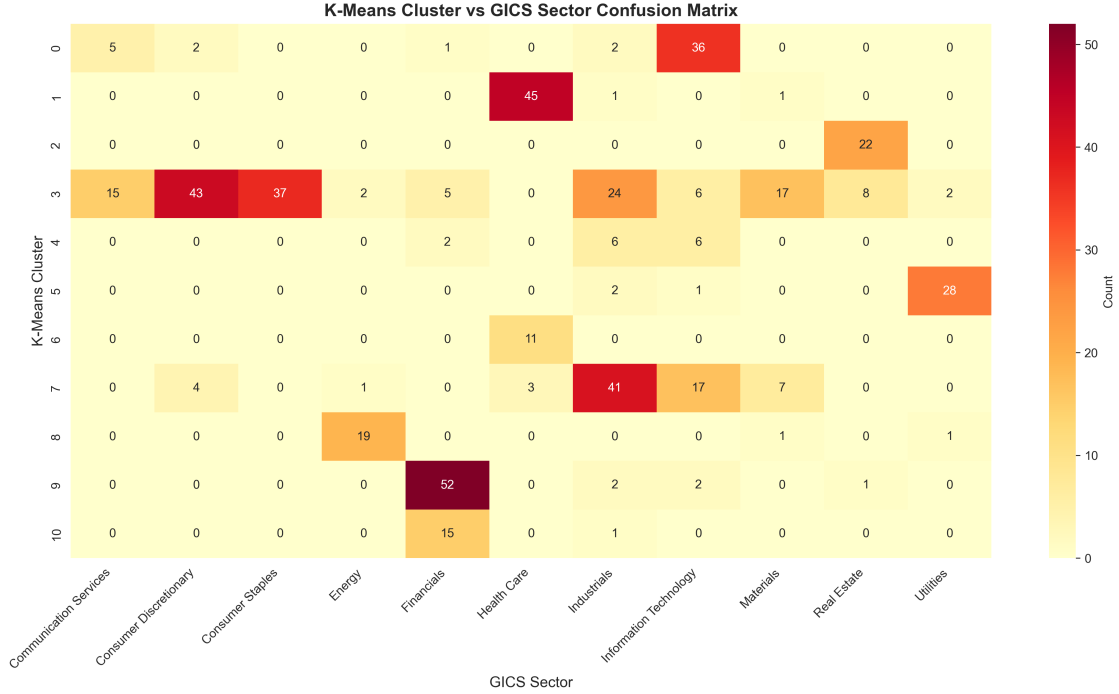


Figure 7: Cluster vs. GICS Sector Confusion Matrix. Each cell (i, j) shows the count of companies that belong to K-Means cluster i and GICS sector j . Dark cells along a single column would indicate perfect alignment between unsupervised clustering and ground truth sector labels. The observed mixing reveals that text-based groupings capture operational and technological characteristics that cut across traditional GICS sector boundaries.

The confusion matrix reveals substantial mixing between clusters and sectors, confirming that text-based groupings differ from GICS classifications. This divergence suggests that descriptions in Wikipedia articles emphasize characteristics that stretch across traditional sector boundaries.

4.3 Supervised Classification Results

4.3.1 Cross-Validation Results

We performed 5-fold stratified cross-validation to evaluate model performance. Table 3 and Figure 8 present the results.

Table 3: 5-Fold Cross-Validation Results (Mean $\pm$ Std)

Classifier	Accuracy
Logistic Regression	0.775 ± 0.026
Linear SVM	0.859 ± 0.014

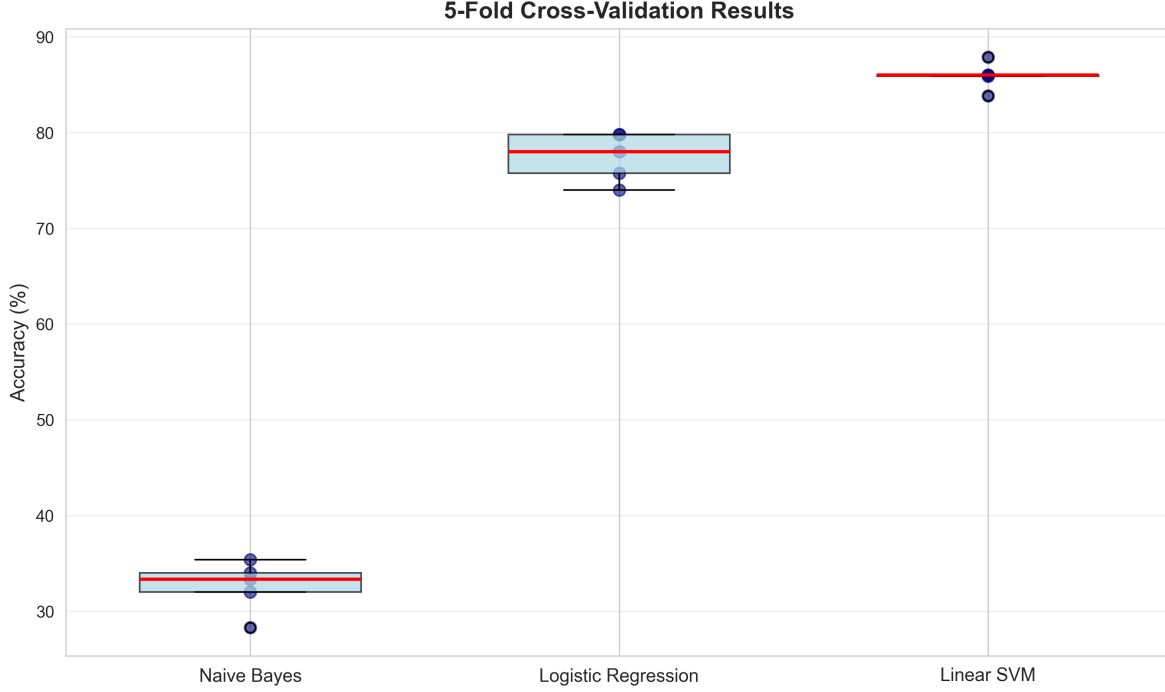


Figure 8: 5-fold cross-validation box plots showing distribution of accuracy across folds. Linear SVM demonstrates consistent performance with low variance, confirming robust generalization.

Cross-validation confirms that Linear SVM achieves stable performance ($85.9\% \pm 1.4\%$). The low standard deviation indicates reliable generalization across different data splits, demonstrating robust model performance.

4.4 Misclassification Analysis

Linear SVM’s cross-validation accuracy of 85.9% corresponds to a 14.1% error rate, indicating that approximately 70 companies across the dataset present classification challenges. Analysis of misclassified companies reveals several patterns:

- **Cross-sector operations:** Companies operating in multiple domains (e.g., technology + finance, healthcare + software)
- **Recent pivots:** Firms that have shifted business models but retain historical GICS classifications
- **Conglomerate diversification:** Diversified corporations whose Wikipedia descriptions emphasize non-primary segments

Table 4 presents five representative misclassification examples that illustrate how Wikipedia’s operational focus can diverge from GICS’s classification criteria.

Ticker	Company	True Sector		Predicted Sector		Rationale
AMZN	Amazon	Consumer	Discretionary	Communication	Services	E-commerce giant; Wikipedia emphasizes cloud services (AWS), digital content (Prime Video), media operations
UBER	Uber Technologies	Industrials		Information	Technology	Ride-sharing platform; Wikipedia focuses on mobile app technology, algorithms, digital marketplace infrastructure
MMM	3M	Industrials		Materials		Diversified manufacturer; Wikipedia extensively covers materials science, adhesives, chemical products
TGT	Target Corporation	Consumer	Staples	Consumer	Discretionary	Retail chain; Wikipedia emphasizes discretionary merchandise, fashion collaborations, shopping experience
FSLR	First Solar	Information	Technology	Utilities		Solar panel manufacturer; Wikipedia describes renewable energy generation, power plant operations

Table 4: Representative misclassification case studies from Linear SVM showing companies that challenge traditional sector boundaries. These examples demonstrate how Wikipedia’s emphasis on operational characteristics can differ from GICS’s classification criteria.

These case studies reveal a consistent pattern: companies with significant technology components (Amazon, Uber), conglomerate diversification (3M), or complex operational models (First Solar) are often predicted to belong to sectors that emphasize their Wikipedia-documented characteristics rather than their primary revenue source. This suggests that text-based classification captures *how companies operate* whereas GICS captures *what they sell*.

4.5 TF-IDF Interpretability: Cross-Sector Clusters

To provide transparency into what unsupervised clustering discovered, we analyze Spectral clustering’s thematic structure. Rather than post-hoc alignment with GICS, we examine clusters directly based on their sector composition and keywords. This reveals two patterns: (1) *pure single-sector clusters* that validate GICS taxonomy, and (2) *cross-sector clusters* that reveal hidden operational themes spanning multiple GICS sectors.

4.5.1 Cross-Sector Themes

Table 5 presents clusters where no single GICS sector dominates (>70%), revealing operational themes that transcend traditional sector boundaries.

Cluster Theme	Keywords	Sector Distribution
Entertainment/ Media/Telecom (n=30)	entertainment, resorts, hotels, communications, studios, media, network, towers	Communication Services (50%), Consumer Discretionary (27%), Real Estate (17%)
Retail/Distribution (n=122)	products, stores, brands, business, food, sales, largest, united states	Consumer Staples (30%), Consumer Discretionary (26%), Industrials (18%), Materials (12%) + 6 others
Industrial Manufacturing (n=62)	industrial, systems, aerospace, manufacturing, equipment, technologies, production	Industrials (69%), Materials (15%), Consumer Discretionary (7%), IT (7%)

Table 5: Cross-sector clusters where no single GICS sector dominates. The Retail/Distribution cluster (n=122) is the largest, spanning 10 different GICS sectors—demonstrating that operational characteristics around consumer products and distribution cut across traditional sector boundaries. The Entertainment/Media/Telecom cluster unites communication infrastructure (towers) with content creation (studios) and hospitality (resorts), revealing how digital convergence blurs sector lines.

These cross-sector clusters suggest that modern companies increasingly defy single-sector categorization—a pattern we explore further in Section 6.2. For instance, American Tower (Real Estate), AT&T (Communication Services), and Charter Communications (Communication Services) cluster together based on shared telecommunications infrastructure vocabulary, despite GICS placing them in different sectors.

4.5.2 Pure Single-Sector Clusters

In contrast, Table 6 shows clusters with >90% purity from a single GICS sector, demonstrating that some operational domains align perfectly with GICS taxonomy.

Dominant Sector	Keywords	Purity
Real Estate	real estate, estate, properties, investment trust, apartment, reit	100% (22/22)
Utilities	utility, energy, electric, gas, power, electricity, natural gas, nuclear	97% (30/31)
Energy	basin, gas, natural gas, oil, petroleum, permian, barrels, oil gas	95% (19/20)
Health Care - Insurance	medicare, healthcare, care, health, patients, medicaid, hospital	100% (10/10)
Health Care - Pharma	medical, fda, drug, pharmaceutical, clinical, treatment, diagnostics	96% (46/48)
Financials - Insurance	insurance, casualty, reinsurance, property casualty, risk, life insurance	95% (18/19)

Table 6: Pure single-sector clusters validate GICS taxonomy for operationally distinct domains. Notably, Health Care splits into two sub-themes (insurance-focused vs pharmaceuticals), and Financials concentrates on insurance rather than banking, suggesting that unsupervised clustering discovers finer-grained operational distinctions than the 11-sector GICS taxonomy.

The pure clusters demonstrate that TF-IDF features successfully capture domain-specific terminology: energy companies use geological vocabulary (basin, permian), utilities emphasize infrastructure (electric, power, nuclear), and pharmaceuticals focus on regulatory processes (FDA, clinical trials). The 95-100% purity rates confirm that textual descriptions naturally separate companies into coherent operational categories when those categories represent truly distinct business models.

5 Potential Application: Text-Based Factor Analysis

5.1 Motivation: Continuous Structure, Not Discrete Clusters

The uniformly low silhouette scores (<0.05) across all values of k (Figure 3) reveal a fundamental characteristic of the data: *companies do not form discrete clusters in textual space*. This finding is not a limitation of our clustering approach—rather, it demonstrates that operational characteristics exist along continuous spectrums.

This observation connects to *factor investing* [Fama and French, 1993], where assets are characterized by continuous factor exposures (e.g., value, growth, momentum) rather than discrete categories. Extending this framework to text, companies could be positioned along operational dimensions (digital transformation, innovation intensity, globalization) that transcend traditional sector boundaries.

5.2 Sector Factor Profiles

We demonstrate the feasibility of text-based factors by defining 8 operational dimensions through keyword aggregation. For each factor, we manually selected 10-15 semantically related terms (e.g., *Digital Transformation*: digital, cloud, platform, software, AI, automation; *Innovation R&D*: research, development, innovation, patent, laboratory) and summed their TF-IDF scores across matched keywords.

Percentile ranking for factor differentiation: To enable factor-based portfolio construction, we convert raw TF-IDF sums to percentile ranks (0-100 scale). This approach preserves relative ordering while creating uniform distribution across the full range—critical for forming quintile portfolios in factor investing. Percentile ranking yields high variation ($\text{std} \approx 28-29$), enabling meaningful differentiation between high- and low-factor companies.

We then computed sector-level profiles by averaging company percentile scores within each GICS sector. Figure 9 presents average factor scores across GICS sectors.



Figure 9: Average text-based factor scores by GICS sector (radar chart). Information Technology leads on Digital Transformation (84.8), Healthcare on Healthcare Medical (92.2) and Innovation R&D (67.2), while Utilities specialize in Sustainability ESG (91.8). The percentile-ranked scores create clear sector differentiation, with each sector exhibiting 1-2 specialized dimensions where they score substantially above average (>70), enabling factor-based portfolio construction.

The profiles confirm sector specializations align with domain expectations: Information Technology scores highest on Digital Transformation (84.8), Healthcare on Healthcare Medical (92.2), Utilities on Sustainability ESG (91.8), and Materials on Industrials Manufacturing (78.7). The percentile ranking methodology successfully differentiates sectors: only 10% of companies score in the narrow 45-55 range, with most distributed across the full 0-100 spectrum. This high variation validates that text-based factors can capture meaningful operational differences and enables the quintile-based portfolio construction required for factor investing strategies.

5.3 Implications and Future Work

The percentile ranking methodology successfully enables factor-based portfolio construction. With standard deviation of approximately 28-29, companies exhibit sufficient variation to form meaningful quintile portfolios. For example, the Digital Transformation factor can separate companies into five groups of approximately 100 companies each, enabling traditional long-short factor strategies (long top quintile, short bottom quintile).

This framework addresses a fundamental limitation of discrete sector classifications: companies exist along continuous operational spectrums rather than in distinct categorical buckets. Text-based factors capture this continuity while preserving interpretability through keyword-based construction.

Practical application: Rather than discrete sector allocation (“20% Technology, 30% Health-

care”), investors could target continuous factor exposures (“companies scoring >80 on Digital Transformation”), potentially capturing EPAM Systems (Information Technology), T-Mobile (Communication Services), and CoStar Group (Real Estate) together based on shared operational characteristics invisible to GICS-based analysis.

Validation requirements: Before deploying this approach in practice, systematic validation is required:

- **Factor pricing tests:** Determine whether text-based factors predict returns or risk through Fama-MacBeth regressions [Fama and French, 1993]. If these factors are not priced in equity markets, they provide purely descriptive rather than predictive value.
- **Portfolio backtesting:** Construct long-short quintile portfolios for each factor and evaluate risk-adjusted performance against sector-based benchmarks over historical periods.
- **Stability analysis:** Track how factor scores evolve as Wikipedia content updates. Stable factors should exhibit gradual evolution reflecting genuine business changes rather than volatile swings from editorial revisions.
- **Alternative data sources:** Compare factor scores derived from Wikipedia against those from SEC 10-K filings or earnings call transcripts to validate robustness across data sources.
- **Factor orthogonality:** Assess whether the 8 manually-defined factors are sufficiently independent, or whether dimensionality reduction (e.g., PCA on factor scores) reveals fewer underlying dimensions.

The current analysis demonstrates technical feasibility: percentile-ranked text-based factors create sufficient variation for quintile formation and exhibit interpretable sector specializations. However, economic validation—demonstrating that these factors predict returns or improve risk management—remains future work.

6 Discussion

6.1 Interpretation of Results

The stark contrast between high supervised classification accuracy (85.9% cross-validated) and moderate unsupervised clustering agreement (ARI: 0.336–0.439, NMI: 0.586–0.626) reveals a fundamental insight: *Wikipedia text contains information about GICS sectors, but when companies are grouped without sector labels, alternative organizational structures emerge.*

This suggests two possibilities:

1. **GICS captures only one aspect:** Traditional sectors may represent historical industry groupings rather than operational or technological similarity
2. **Textual descriptions reveal latent themes:** Wikipedia articles emphasize cross-cutting themes (e.g., “digital transformation,” “data analytics”) that transcend sector boundaries

GICS hierarchical structure: The GICS framework employs a four-level hierarchy: 11 Sectors → 25 Industry Groups → 74 Industries → 163 Sub-Industries. Our analysis operates solely at the coarsest Sector level, where the 163 Sub-Industries provide substantially finer granularity (e.g., separating “Biotechnology” from “Pharmaceuticals”). However, even Sub-Industries remain fundamentally *categorical*—each company receives a single discrete label. Text-based factors offer a

continuous representation where companies exhibit varying degrees of exposure across multiple operational dimensions simultaneously, complementing GICS’s categorical taxonomy with continuous factor exposures suitable for portfolio construction.

The preliminary factor analysis in Section 5 provides evidence for the second interpretation: operational characteristics form continuous dimensions that explain the low silhouette scores and cross-sector similarities.

6.2 Cross-Sector Companies and Emerging Patterns

The moderate NMI (0.626) indicates that discovered clusters partially align with GICS while also capturing alternative groupings. This supports the hypothesis that modern companies increasingly defy single-sector categorization. Companies classified as “Information Technology” under GICS may cluster with “Financials” companies when both emphasize digital platforms and data services.

6.2.1 Inter-Sector Textual Similarity

To quantify cross-sector relationships, we computed pairwise cosine similarity between sector centroids (average TF-IDF feature vectors). Figure 10 presents the resulting similarity matrix.



Figure 10: Inter-sector textual similarity matrix based on TF-IDF feature centroids. Darker green indicates higher cosine similarity. Industrials exhibits high similarity with Information Technology (0.787), Consumer Discretionary (0.786), and Materials (0.752), suggesting convergence around manufacturing technology and supply chain operations.

The analysis reveals several noteworthy patterns:

1. **Industrials as a Hub Sector:** Industrials shows high textual similarity with IT (0.787), Consumer Discretionary (0.786), Materials (0.752), and Health Care (0.683). This suggests that industrial companies' Wikipedia articles emphasize technology adoption, supply chain complexity, and cross-sector applications.
2. **IT-Financials Convergence:** Information Technology and Financials exhibit moderate similarity, reflecting the rise of FinTech and digital banking platforms that blur sector boundaries.
3. **Consumer Sector Overlap:** Consumer Discretionary and Consumer Staples share substantial similarity (0.814), indicating that retail, distribution, and consumer-facing business language transcends product durability categories.
4. **Distinct Sectors:** Energy, Utilities, and Real Estate show lower cross-sector similarity, confirming their specialized vocabulary and regulatory focus.

These quantitative similarity scores corroborate the earlier finding that text-based representations capture operational themes that span traditional sector boundaries. The prominence of Industrials in multiple high-similarity pairs reflects the sector’s diversity and its role as a bridge between manufacturing, technology, and consumer markets.

6.3 Implications for Market Analysis

The 14.1% misclassification rate from cross-validated Linear SVM suggests that approximately one-seventh of companies exhibit textual characteristics inconsistent with their GICS sector. For investors and analysts, this finding highlights:

- **Hidden correlations:** Companies in different GICS sectors may face similar risks if they share operational themes. Text-based factor analysis (Section 5) reveals that sectors overlap on 6-7 out of 8 operational dimensions, suggesting traditional sector-based risk models might miss significant cross-sector correlations.
- **Sector reclassification candidates:** Misclassified companies may benefit from updated sector assignments or represent firms genuinely operating across multiple domains.
- **Alternative data for alpha:** Text-based analysis could complement traditional sector studies, particularly for identifying change in operational activities before they manifest in official documentations.

6.4 Limitations

- **Data Source:** Wikipedia articles reflect public perception and may not capture internal business structure or recent developments
- **Information Leakage:** Wikipedia articles may explicitly mention a company’s GICS sector or use sector-specific terminology in ways that facilitate classification without capturing operational characteristics. However, the unsupervised clustering results (which do not use sector labels) achieve moderate alignment with GICS (NMI: 0.626), suggesting that Wikipedia text captures genuine operational themes beyond direct sector mentions
- **Uncaptured Temporal Dynamics:** Company’s business models evolve over time, yet this analysis represents only a single time snapshot
- **Text Quality Variance:** Article length varied significantly (median: 1,579 words; mean: 2,448 words), potentially biasing results toward larger, more prominent companies
- **Variance Retention:** Achieving only 78% variance with 300 components suggests that text data is highly sparse
- **GICS Evolution:** GICS itself undergoes periodic revisions (e.g. major framework revamp in 2023 [S&P Dow Jones Indices, 2025]) to accommodate market evolution. However, institutional classification frameworks face inherent tension between stability and responsiveness. This structural constraint means GICS classifications may lag operational reality, particularly for companies undergoing digital transformation or cross-sector convergence
- **Preliminary factor framework:** The text-based factors in Section 5 are exploratory and require systematic validation before deployment in portfolio construction

7 Conclusion

This project demonstrates that text-based features from Wikipedia articles contain meaningful information about S&P 500 company sectors. Linear SVM achieves 85.9% cross-validated accuracy predicting GICS sectors, with sublinear term frequency scaling providing notable improvement. Unsupervised clustering reveals latent structures that both align with and diverge from GICS taxonomy. Spectral clustering achieves the best unsupervised performance (ARI: 0.439, NMI: 0.626), capturing approximately 63% of GICS structure while emphasizing operational and technological themes.

The uniformly low silhouette scores reveal that companies exist in continuous operational space rather than discrete clusters, motivating potential applications in factor-based portfolio construction. The 14.1% of companies that were difficult to classify represent candidates for sector reclassification or evidence of increasing industry convergence.

These findings highlight both the strengths and limitations of static sector classification systems. While GICS provides a valuable organizing framework, text-based analysis reveals alternative groupings that may better reflect modern business realities. Future work should:

- **GICS Sub-Industry alignment:** Investigate whether text-based factors correlate with GICS’s 163 Sub-Industry classifications, potentially revealing which operational dimensions drive finer-grained categorical distinctions
- **Temporal analysis:** Track how companies’ textual representations and cluster assignments evolve over time
- **Alternative text sources:** Incorporate company filings, earnings calls, or related news articles for comparison
- **Deep learning:** Explore transformer-based embeddings or the use of Large Language Models (LLMs) while maintaining interpretability through attention mechanisms or through the use of reasoning models
- **Hierarchical clustering:** Investigate multi-level taxonomies that could capture both traditional sectors and cross-cutting themes
- **Factor validation and backtesting:** Formalize text-based factor definitions, construct factor-mimicking portfolios, and evaluate whether these factors are priced in equity markets through Fama-MacBeth regressions
- **Dynamic factor tracking:** Build pipelines to monitor factor score evolution as Wikipedia content updates, enabling real-time detection of business model transformations

In conclusion, this work suggests that the question is not whether text-based classification can replace GICS, but rather how textual analysis can complement existing framework to provide a more comprehensive understanding of company relationships in an increasingly interconnected and innovation-driven economy. The preliminary factor framework demonstrates one such complementary approach, positioning companies along operational spectrums that may inform portfolio construction and risk management beyond discrete sector boundaries.

References

- Mehdi Allahyari, Seyedamin Pouriyeh, Mehdi Assefi, Saeid Safaei, Elizabeth D Trippe, Juan B Gutierrez, and Krys Kochut. A brief survey of text mining: Classification, clustering and extraction techniques. *arXiv preprint arXiv:1707.02919*, 2017.
- Leland Bybee, Bryan Kelly, and Yinan Su. Narrative asset pricing: Interpretable systematic risk factors from news text. *The Review of Financial Studies*, 36(12):4759–4787, 2023.
- Eugene F Fama and Kenneth R French. Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56, 1993.
- Gerard Hoberg and Gordon Phillips. Text-based network industries and endogenous product differentiation. *Journal of Political Economy*, 124(5):1423–1465, 2016.
- Allen H Huang, Hui Wang, and Yi Yang. Finbert: A large language model for extracting information from financial text. *Contemporary Accounting Research*, 40(2):806–841, 2023.
- Lawrence Hubert and Phipps Arabie. Comparing partitions. *Journal of Classification*, 2(1):193–218, 1985.
- Martin Lamby and Daniel Iseman. Classifying companies by industry using word embeddings. In *Natural Language Processing and Information Systems (NLDB 2018)*, pages 377–388. Springer, 2018.
- Christopher D Manning, Prabhakar Raghavan, and Hinrich Schütze. *Introduction to Information Retrieval*. Cambridge University Press, 2008.
- Gautier Marti. Building a s&p 500 company classification from wikipedia articles (guided by chatgpt). <https://gmarti.gitlab.io/quant/2023/05/07/wikipedia-network-companies-sentence-transformers.html>, May 2023. Accessed: 2025-11-22.
- MSCI. The global industry classification standard (gics). <https://www.msci.com/indexes/index-resources/gics>, 2025. Accessed: 2025-11-22.
- Andrew Ng, Michael Jordan, and Yair Weiss. On spectral clustering: Analysis and an algorithm. In *Advances in Neural Information Processing Systems*, volume 14, 2001.
- Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, et al. Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- Gerard Salton and Christopher Buckley. Term-weighting approaches in automatic text retrieval. *Information Processing & Management*, 24(5):513–523, 1988.
- S&P Dow Jones Indices. Gics: Global industry classification standard. <https://www.spglobal.com/spdji/en/landing/topic/gics/>, 2025. Accessed: 2025-11-22.
- Alexander Strehl and Joydeep Ghosh. Cluster ensembles—a knowledge reuse framework for combining multiple partitions. *Journal of Machine Learning Research*, 3:583–617, 2002.
- Winton Research. Systematic methods for classifying equities. <https://www.belmontinvestments.com/cimg/file/articles/50/pdf/180830wintonequityclassification.pdf>, 2018. Accessed: 2025-11-30.